

TSGS: Improving Gaussian Splatting for Transparent Surface Reconstruction via Normal and De-lighting Priors

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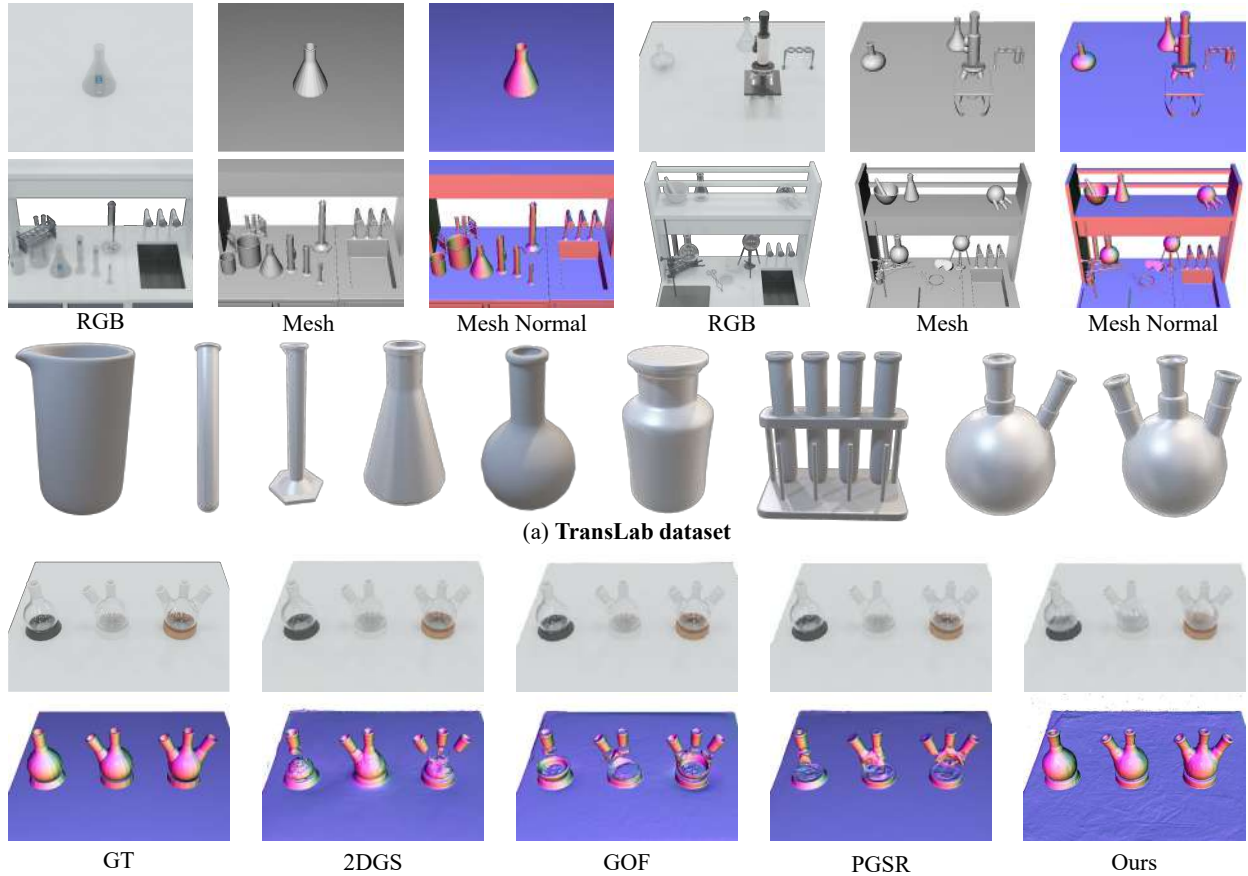


Figure 1: (a) We introduce TransLab, a novel dataset specifically designed for evaluating transparent object reconstruction. TransLab comprises 8 diverse, high-resolution 360° scenes in laboratory settings. It features a wide array of challenging transparent glassware, including test tubes, beakers, volumetric flasks etc. (b) Comparative results on TransLab demonstrate the superior capability of our proposed TSGS method in accurately reconstructing geometry and appearance compared to prior state-of-the-art approaches.

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Abstract

Reconstructing transparent surfaces is essential for tasks such as robotic manipulation in labs, yet it poses a significant challenge for 3D reconstruction techniques like 3D Gaussian Splatting (3DGS). These methods often encounter a transparency-depth dilemma, where the pursuit of photorealistic rendering through standard α -blending undermines geometric precision, resulting in considerable depth estimation errors for transparent materials. To address this issue, we introduce **Transparent Surface Gaussian Splatting (TSGS)**, a new framework that separates geometry learning from appearance refinement. In the geometry learning stage, TSGS focuses on geometry by using specular-suppressed inputs to accurately represent surfaces. In the second stage, TSGS improves visual fidelity through anisotropic specular modeling, crucially maintaining the established opacity to ensure geometric accuracy. To enhance depth inference, TSGS employs a first-surface depth extraction method. This technique uses a sliding window over α -blending weights to pinpoint the most likely surface location and calculates a robust weighted average depth. To evaluate the transparent surface reconstruction task under realistic conditions, we collect a **TransLab** dataset that includes complex transparent laboratory glassware. Extensive experiments on TransLab show that TSGS achieves accurate geometric reconstruction and realistic rendering of transparent objects simultaneously within the efficient 3DGS framework. Specifically, TSGS significantly surpasses current leading methods, achieving a 37.3% reduction in chamfer distance and an 8.0% improvement in F1 score compared to the top baseline. Additionally, TSGS maintains high-quality novel view synthesis, evidenced by a 0.41dB gain in PSNR, demonstrating that TSGS overcomes the transparency-depth dilemma. The code and dataset are available at <https://longxiang-ai.github.io/TSGS/>.

CCS Concepts

• Computing methodologies → Rendering; Shape modeling.

Keywords

3D Gaussian Splatting; Surface Reconstruction; Mesh Reconstruction; Transparency Reconstruction; Novel View Synthesis

1 Introduction

Novel view synthesis and geometry reconstruction are key tasks in computer vision, essential for AR/VR [17, 77], 3D content creation [13, 23, 55, 63], and autonomous driving. These applications require high realism and detailed geometry. Techniques like Neural Radiance Fields (NeRF) [26, 48, 49, 76] have advanced high-fidelity results [2, 5, 41, 52, 67]. Recently, 3D Gaussian Splatting (3DGS) [31] has emerged as a significant advancement by using explicit 3D Gaussians and an efficient, differentiable rasterizer to enable fast training and real-time rendering. However, 3DGS still struggles with accurately reconstructing challenging materials, especially transparent surfaces.

This limitation in reconstructing transparent surfaces is crucial, especially for applications like robotic laboratory systems that require millimeter-precise manipulation of glassware, such as beakers and test tubes. The main issue with 3DGS is its inability to accurately handle transparent surfaces. This is due to a fundamental

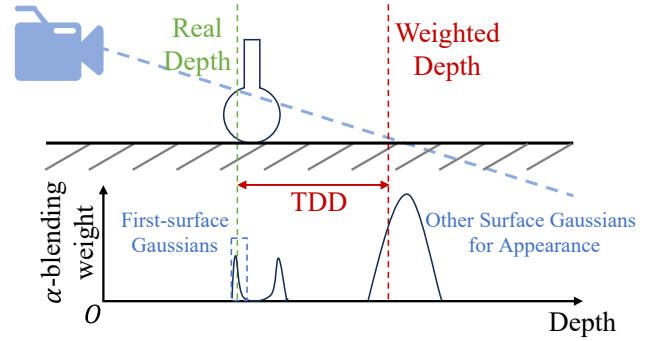


Figure 2: The transparency-depth dilemma in Gaussian Splatting (§1): Previous GS-based methods [14, 29] typically derive depth maps using standard α -blending (Eq. 1, 4), averaging depths of Gaussians along the ray weighted by their appearance-optimized weights. For transparent objects, this blending inherently mixes contributions from the first surface and transmitted background details, failing to accurately localize the true first-surface depth.

conflict: the model is optimized primarily for visual appearance, not for accurately deriving depth using standard α -blending techniques. We refer to this problem as the **transparency-depth dilemma**.

Specifically, as shown in Fig. 2, traditional 3D Gaussian Splatting methods [9, 25] render depth using α -blending, where depth values are derived from weighted averages based on the opacity of contributing Gaussians. However, this opacity is mainly optimized for visual appearance rather than accurate depth measurement. While effective for opaque objects—where high-opacity Gaussians cluster near the surface—it fails for transparent materials, giving rise to the transparency-depth dilemma.

The main problem stems from the conflicting goals of modeling appearance and achieving geometric accuracy. In 3DGS, to depict visual transparency and show objects behind a surface, the model reduces the opacity of Gaussians that represent the transparent surface. Consequently, when these appearance-optimized opacities are used to weight depth rendering, the actual transparent surface’s contribution is minimized. This causes deeper background Gaussians to disproportionately affect the weighted average, leading to rendered depths that are systematically greater than the actual physical depths. This results in depth maps that inaccurately depict transparent regions as being farther away than they really are, compromising precise 3D reconstruction.

To address the transparency-depth dilemma, a simple solution might be to abandon α -blending and record the depth where a ray first intersects any Gaussian, known as “nearest depth”. However, this method is flawed because it depends on the properties of individual Gaussian primitives, which do not have stable geometric meaning alone—only the combined effect of Gaussian clusters creates stable surfaces in 3DGS. This leads to two problems. 1) Systematic underestimation bias: In 3DGS, Gaussians are distributed around the true surface. While the surface depth is derived from the collective Gaussian cluster, using the nearest intersection depth typically captures Gaussians closer to the camera than the actual

surface. This consistently results in depth values that are smaller (closer) than the true physical depth. 2) Instability and noise artifacts: Since optimization focuses on overall appearance, the precise location and shape of any single Gaussian are not strictly controlled. Therefore, the specific Gaussian that causes the nearest intersection can vary greatly, particularly near object boundaries and thin structures. This leads to noisy and unreliable depth estimations.

To address the outlined challenges, we propose a novel two-stage training framework that separates geometry and appearance optimization, complemented by a first-surface extraction method. In the first stage, we use normal maps and de-lighted images, which render transparent regions as darker and less transparent, to create geometrically accurate Gaussian distributions. In the second stage, we fix the opacity parameters of the Gaussians and enhance the model’s ability to depict complex transparent appearances using anisotropic spherical Gaussian (ASG) components. This approach maintains the accurate geometry established in the first stage while enabling photorealistic rendering of transparency. At inference time, we introduce a physically principled first-surface extraction technique that accurately pinpoints the initial surface intersection for each ray. This method overcomes the transparency-depth dilemma and avoids the noise artifacts typical of simpler nearest-depth approaches.

Widely used datasets [4, 27, 34] lack specialized focus on transparent objects with complex geometries like laboratory glassware, yet the growing field of embodied AI for automated labs demands precise reconstruction and manipulation capabilities for these transparent items. To address this gap, we collect the TransLab dataset specifically designed to evaluate transparent surface reconstruction. This dataset includes 8 different 360° scenes of common laboratory environments with transparent equipment like test tubes, beakers, and flasks. Our experimental results show significant improvements over existing methods, especially the strong PGSR baseline. With TransLab, our TSGS method reduced the chamber distance by 37.3% and improved the F1 score for geometric precision by 8.0%. It also enhanced visual quality, increasing PSNR by +0.41dB. Additionally, TSGS achieved top geometric accuracy on the opaque DTU dataset [27], demonstrating the robustness and versatility of our approach.

Overall, our contributions can be summarized as follows:

- A two-stage training strategy that initially sets up accurate geometry through specular-decoupled learning, followed by appearance refinement while maintaining geometric integrity. This approach ensures robust reconstruction of transparent objects.
- A first-surface depth extraction technique that achieves high accuracy in transparent surface reconstruction through a maximum-weight window aggregation algorithm.
- A TransLab dataset that features transparent laboratory equipment in realistic environments. Extensive experiments show significant improvements of our method over current state-of-the-art methods.

2 Related Work

Traditional Surface Reconstruction. Early methods operated on unstructured 3D point clouds, utilizing techniques such as Poisson reconstruction [30], Delaunay triangulation [6], Alpha shapes [19],

and MLS [36]. Alternatively, volumetric methods represented geometry via discrete 3D fields, as seen in Marching Cubes [45], level-set methods [83], and signed distance fields (SDFs) [16]. Multi-view geometry techniques leveraged camera information to recover 3D structure, including SfM [60], MVS [21], and Shape-from-Shading [82]. Many multi-view pipelines involved intermediate representations like point clouds [37], volumes [35], or depth maps [61], establishing dense correspondences via patch-based matching [1] before generating surfaces through triangulation [8] or implicit fitting [30]. Despite widespread adoption, these traditional methods are sensitive to noise and struggle with transparent or reflective surfaces that violate photometric assumptions. Recent works improve robustness by integrating deep learning into matching [59, 65]. In this work, we assume known poses (e.g., via SLAM [7, 10, 11] or SfM [51, 68]) and focus on surface reconstruction under fixed camera parameters.

Neural Surface Reconstruction. Neural methods have advanced surface reconstruction by leveraging learned priors and differentiable rendering. Early approaches attempted to directly predict 3D shapes from images, representing geometry as point clouds [20, 42], voxels [15, 71], meshes [40, 66], or implicit fields [47, 54]. While enabling end-to-end training, these often require extensive 3D supervision and incur high computational costs. A significant breakthrough came with NeRF [48], which represent scenes as continuous 5D functions. Subsequent variants like Mip-NeRF [2] and Zip-NeRF [5] improved rendering quality, while others like NeuS [67] and VolSDF [74] adapted NeRF specifically for high-fidelity surface reconstruction using SDF-based representations. To address the computational bottlenecks of purely MLP-based models, accelerated representations have been developed, decomposing scenes into structures like points [72], voxels [43], or hybrid combinations [39, 41], thereby improving efficiency without sacrificing reconstruction quality. Several methods [18, 22, 46, 62, 64, 70, 80] have been proposed for transparent object reconstruction. These approaches often leverage NeRF-based techniques that model physical effects like refraction through volumetric ray integration, but this leads to lengthy training times and slow rendering speeds.

Gaussian Splatting-based Surface Reconstruction. A recent paradigm shift occurred with 3DGS [31], enabling real-time rendering and fast training via explicit 3D Gaussians. Extracting consistent surfaces from this representation has been an active area of research. Initial work like SuGaR [24] focused on mesh extraction through Poisson reconstruction. Subsequent efforts aimed to enhance geometric quality directly within the Gaussian framework like 2DGS [25], GOF [79], and PGSR [9]. While these methods improve reconstruction for many scenarios, accurately capturing **transparent surfaces** presents specific nuances. Some approaches [32, 38] employ diffusion priors [75, 84] to aid reconstruction, whereas recent studies [50, 69] incorporate ray tracing at the cost of substantial computational overhead. A key observation is that the standard α -blending mechanism, central to 3DGS rendering, creates an inherent challenge for transparent object geometry: the transparency-depth dilemma. While optimizing opacity primarily for appearance successfully enhances visual fidelity, it can lead to inaccuracies when directly using the resulting weighted average for depth calculation. Recognizing this conflict, our work specifically targets the transparency-depth dilemma by proposing

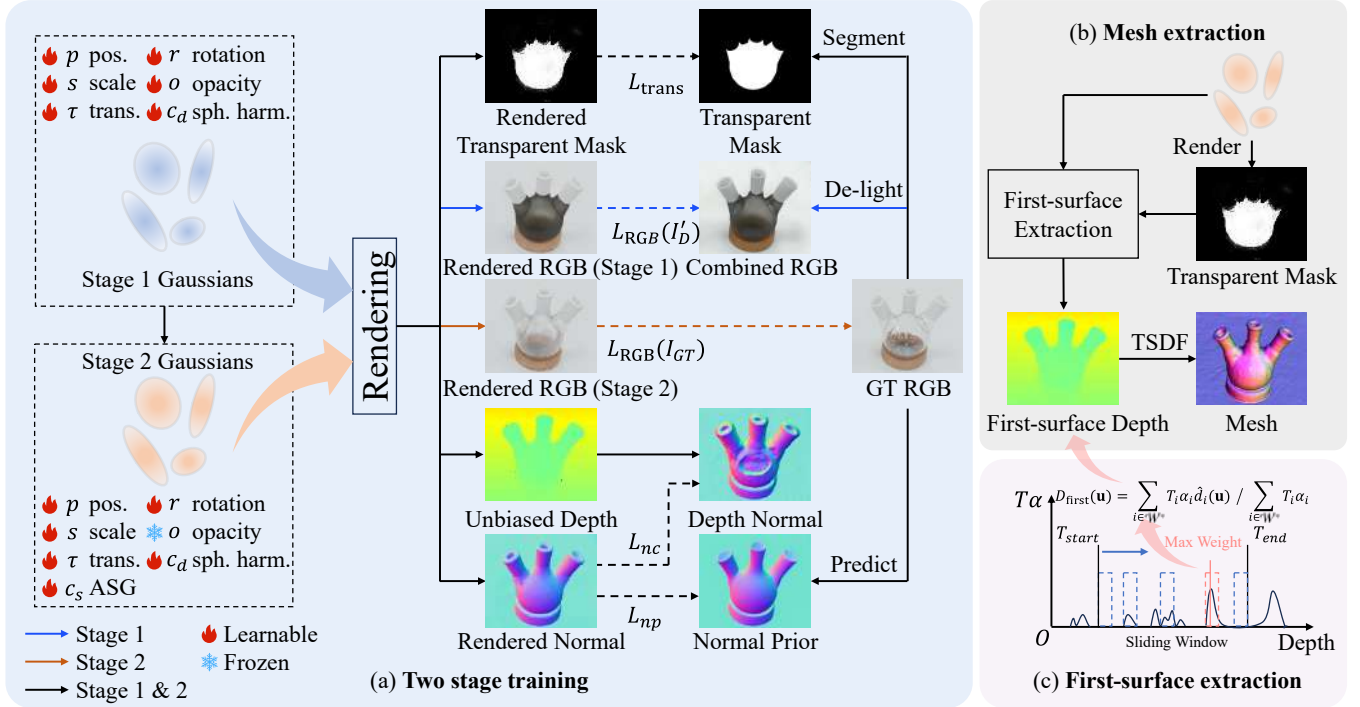


Figure 3: Pipeline of TSGS. (a) The two-stage training process. In Stage 1, 3D Gaussians are optimized using geometric priors and de-lighted image inputs (§3.2). In Stage 2, we fix the opacity parameters and optimize the 3DGS using GT images with specular highlights (§3.3). (b) The inference process utilizes the trained model to extract the first-surface depth map, which is then used for mesh reconstruction (§3.4). (c) Details of the first-surface depth extraction module, which employs a sliding window to find the maximum-weight window along the ray and computes depth via weighted averaging within this window.

a novel approach for accurate first-surface depth extraction from the standard, appearance-optimized Gaussians, aiming to improve geometric reconstruction for transparent objects.

3 Method

We introduce TSGS to tackle the transparency-depth dilemma in transparent surface reconstruction, where photorealistic rendering conflicts with accurate first-surface geometry. As shown in Fig. 3, TSGS employs a three-stage strategy: 1) **Geometry-Focused Learning** (§3.2): learns accurate geometry using de-lighted images [75] and normal priors to suppress specular interference. 2) **Geometry-Preserving Appearance Refinement** (§3.3): refines appearance with original images and Anisotropic Spherical Gaussians (ASG) while preserving geometry. 3) **First-Surface Depth Extraction** (§3.4): extracts accurate first-surface depth using an accumulated transmittance sliding window algorithm during inference. This staged approach yields reconstructions that are both geometrically accurate and visually realistic.

3.1 Preliminary

Gaussian Splatting. 3DGS [31] represents scenes using a set of 3D Gaussians $\{\mathcal{G}_i\}$. Each Gaussian $\mathcal{G}_i$ is parameterized by its center $\mathbf{p}_i$, opacity o_i , color $\mathbf{c}_i$, rotation $\mathbf{r}_i$, and scale $\mathbf{s}_i$. These parameters define the 3D Gaussian distribution in world coordinates as $\mathcal{G}_i(\mathbf{x}) = e^{-\frac{1}{2}(\mathbf{x}-\mathbf{p}_i)^T \Sigma_i^{-1}(\mathbf{x}-\mathbf{p}_i)}$, where the covariance Σ_i is derived from $\mathbf{r}_i$ and $\mathbf{s}_i$. These 3D Gaussians are then projected onto

the 2D image plane. Given the camera extrinsics $\mathbf{W}$ and intrinsics $\mathbf{K}$, the 3D center $\mathbf{p}_i$ and covariance Σ_i are projected to 2D screen coordinates $\mathbf{p}'_i$ and covariance Σ'_i via $\mathbf{p}'_i = \mathbf{K}\mathbf{W}[\mathbf{p}_i, 1]^T$ and $\Sigma'_i = \mathbf{J}\mathbf{W}\Sigma_i\mathbf{W}^T\mathbf{J}^T$, where $\mathbf{J}$ is the Jacobian of the affine approximation for the projective transformation. The depth z_i of the Gaussian center in camera coordinates is obtained via the camera transformation: $[x_i, y_i, z_i, 1]^T = \mathbf{W}[\mathbf{p}_i, 1]^T$.

Unbiased Depth Rendering. Standard 3DGS renders color $\mathbf{C}$ for each pixel $\mathbf{u}$ via α -blending. Previous methods [14, 29] typically render the depth map D by blending the camera-space depth z_i of each Gaussian using the same α -blending weights:

$$\mathbf{C} = \sum_{i \in N} T_i \alpha_i \mathbf{c}_i, \quad D_{\text{standard}} = \sum_{i \in N} T_i \alpha_i z_i \quad (1)$$

Here, the rendering process involves sorting the Gaussians intersecting the pixel ray by its center depth. For each Gaussian $\mathcal{G}_i$ along the sorted list, $T_i = \prod_{j=1}^{i-1} (1 - \alpha_j)$ represents the accumulated transmittance, indicating the probability that the ray has passed through the preceding Gaussians from 1 to $i-1$ without being absorbed. The term $\alpha_i = o_i \mathcal{G}_i(\mathbf{u} | \mathbf{p}'_i, \Sigma'_i)$ is the opacity contribution of the current Gaussian $\mathcal{G}_i$ at pixel $\mathbf{u}$.

To encourage Gaussians to form planar structures, PGSR [9] introduces a flatten regularization $\mathcal{L}_{\text{flatten}}$ during training:

$$\mathcal{L}_{\text{flatten}} = \sum_i |\min(s_{i,1}, s_{i,2}, s_{i,3})|_1. \quad (2)$$

This allows approximating the normal $\mathbf{n}_i$ of the Gaussian $\mathcal{G}_i$ from the axis of the smallest scale component. Building on this, PGSR [9] introduces an unbiased depth rendering method as an alternative to D_{standard} . First, it renders both a world-space normal map $\mathbf{N}_w$ (blending Gaussian normals $\mathbf{n}_i$) and a distance map $\mathcal{D}$ (blending per-Gaussian distances d_i from the camera center to the Gaussian plane):

$$\mathbf{N}_w = \sum_{i \in N} T_i \alpha_i \mathbf{n}_i, \quad \mathcal{D} = \sum_{i \in N} T_i \alpha_i d_i. \quad (3)$$

Second, the unbiased depth map $D_{\text{unbiased}}(\mathbf{u})$ for a pixel $\mathbf{u}$ is derived by the ray direction $\mathbf{v}_u$, the distance $\mathcal{D}(\mathbf{u})$ from the plane and the plane normal $\mathbf{N}_w(\mathbf{u})$ in the world coordinate system:

$$D_{\text{unbiased}}(\mathbf{u}) = \frac{\mathcal{D}(\mathbf{u})}{\mathbf{N}_w(\mathbf{u}) \cdot \mathbf{v}_u}. \quad (4)$$

While the unbiased depth D_{unbiased} offers improvements over the standard blended depth D_{standard} , it ultimately derives depth from quantities ($\mathcal{D}$ and $\mathbf{N}_w$) obtained via standard α -blending. This process inherently weights contributions from Gaussians in all surfaces intersected along the ray, using weights optimized primarily for appearance. Such averaging across depth makes it challenging to accurately isolate the first surface for transparent objects, reflecting the core transparency-depth dilemma and motivating our alternative first-surface extraction approach detailed in §3.4.

3.2 Stage 1: Geometry-Focused Learning

Pre-processing. Reconstructing the geometry of transparent objects solely from RGB images is inherently **ill-posed** due to the ambiguity caused by complex light condition. To address this, we leverage strong geometric priors obtained from large-scale diffusion models. Specifically, we utilize normal maps $\{N_p\}$ generated by StableNormal [75] and de-lighted image sets $\{I_D\}$ pre-processed by StableDelight [75]. To identify the transparent regions, we employ the Grounded-SAM model [28, 33, 44, 56–58] to segment the transparent parts across the dataset, obtaining transparency masks $\{M_T\}$. Using the transparency mask M_T , we blend the de-lighted image I_D with the original ground truth image I_{GT} to create a hybrid image I'_D , where only the transparent regions identified by M_T are de-lighted (i.e., specular components are suppressed):

$$I'_D = M_T \odot I_D + (1 - M_T) \odot I_{GT} \quad (5)$$

Transparency Attribute Learning. To enhance the model’s understanding of transparency, we introduce a learnable attribute $\tau_i \in [0, 1]$ for each Gaussian $\mathcal{G}_i$, indicating if it represents a transparent ($\tau_i \geq 0.5$) or opaque ($\tau_i < 0.5$) region. To supervise this attribute, we set a transparency threshold θ_T and render a predicted transparency mask by selecting the transparency attribute of the Gaussian whose accumulated transmittance T_i is just above this threshold: $\hat{M}_T(\mathbf{u}) = \tau_j$, where $j = \text{argmax}_i \{T_i \geq \theta_T\}$. We then enforce consistency between $\hat{M}_T$ and the ground truth mask M_T using a Binary Cross-Entropy (BCE) loss:

$$\mathcal{L}_{\text{trans}} = \text{BCE}(\hat{M}_T, M_T). \quad (6)$$

Appearance Modeling. We primarily focus on learning accurate geometry rather than complex appearance details like specular highlights in Stage 1. Therefore, the appearance $\mathbf{c}_{i,1}$ for Stage 1 is represented solely by its diffuse color component $\mathbf{c}_{i,d}$, modeled

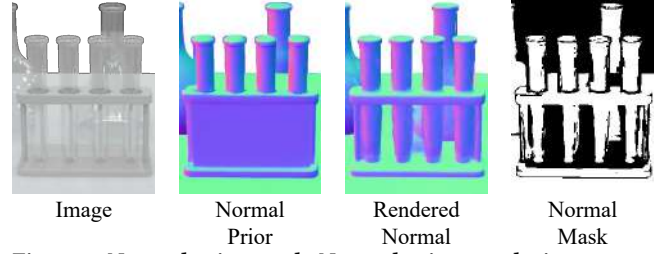


Figure 4: Normal prior mask. Normal priors can be inaccurate for challenging views, potentially misleading the rendered normal. Our normal mask filters out these unreliable priors during loss computation, enhancing the robustness of our normal rendering. (§3.2)

using standard Spherical Harmonics (SH) [31]. This diffuse component is learned by optimizing the appearance loss $\mathcal{L}_{s1}$ against the hybrid de-lighted images I'_D :

$$\mathcal{L}_{s1} = \mathcal{L}_{\text{RGB}}(\hat{I}, I'_D) = (1 - \lambda_r) \|\hat{I}'_D - \hat{I}\|_1 + \lambda_r (1 - \text{SSIM}(I'_D, \hat{I})) \quad (7)$$

where $\lambda_r = 0.2$ balances the L1 and SSIM contributions.

Normal Regularization. As highlighted in the pre-processing (§3.2), reconstructing transparent object geometry solely from RGB images is an ill-posed problem. To mitigate ambiguity and provide strong geometric guidance, especially in textureless or highly reflective areas, we incorporate normal supervision into training. **Normal Prior Loss ($\mathcal{L}_{np}$):** To leverage external geometric information, we enforce consistency between the rendered normals $\mathbf{N}_r$ and the prior normal map $\mathbf{N}_p$ obtained from StableNormal [75]. However, predicted priors $\mathbf{N}_p$ can sometimes be inaccurate due to complex lights. To prevent incorrect priors from degrading the learned geometry, we employ a robust masking strategy, as illustrated in Fig. 4. The loss is calculated as:

$$\mathcal{L}_{np} = \sum M_n \cdot (1 - \mathbf{N}_p^T \mathbf{N}_r). \quad (8)$$

Here, the mask $M_n = (\mathbf{N}_p^T \mathbf{N}_r \geq \theta_n)$, with a threshold θ_n , effectively ignores the prior loss contribution for pixels where the rendered normal $\mathbf{N}_r$ significantly deviates from the prior $\mathbf{N}_p$.

Depth-Normal Consistency Loss ($\mathcal{L}_{nc}$): Following previous practices [3, 31], this loss encourages internal geometric consistency by enforcing agreement between normals computed from the depth map $\mathbf{N}_d$ and the directly rendered normals $\mathbf{N}_r$:

$$\mathcal{L}_{nc} = \sum (1 - \mathbf{N}_d^T \mathbf{N}_r). \quad (9)$$

Thus, the normal regularization term is defined as:

$$\mathcal{L}_n = \mathcal{L}_{np} + \mathcal{L}_{nc}. \quad (10)$$

Total Loss for Stage 1. The final training objective for Stage 1, $\mathcal{L}_{\text{total}, s1}$, combines the stage-specific appearance loss ($\mathcal{L}_{s1}$) with the transparency loss ($\mathcal{L}_{\text{trans}}$), the combined normal loss ($\mathcal{L}_n$), and the flatten regularization loss $\mathcal{L}_{\text{flatten}}$ (defined in Eq. 2):

$$\mathcal{L}_{\text{total}, s1} = \mathcal{L}_{s1} + \lambda_t \mathcal{L}_{\text{trans}} + \lambda_n \mathcal{L}_n + \lambda_f \mathcal{L}_{\text{flatten}} \quad (11)$$

where $\lambda_t = 0.1$, $\lambda_n = 0.1$, and $\lambda_f = 100$ are hyperparameters balancing the contributions of the respective regularization terms during this geometry-focused stage.

3.3 Stage 2: Appearance Refinement

Having established a reliable geometry foundation in Stage 1, our focus in Stage 2 shifts to enhancing the appearance quality, particularly capturing the complex specular and transparent effects present in the original images I_{GT} . Instead of using the hybrid de-lighted images I'_D from Stage 1, we supervise with the original ground truth images I_{GT} . Moreover, to preserve the learned geometric structure represented in the opacity distribution while optimizing for more complex appearance, we **fix** the opacity parameter o_i , preventing incorrect updates driven by the appearance loss against I_{GT} .

Appearance Modeling. To effectively model the strong view-dependency of specular highlights and transmission effects common in transparent objects, we utilize the appearance modeling strategy from Spec-Gaussian [73] to model the specular component. Specifically, we employ anisotropic spherical Gaussians (ASGs) to model the latent feature of specular component $\mathbf{c}_{i,s}$:

$$\mathbf{c}_{i,s} = \Psi(\kappa_i, \gamma(\mathbf{d}_i), \mathbf{n}_i, -\mathbf{d}_i) \quad (12)$$

where Ψ is a tiny MLP decoder, κ_i represents the latent feature derived from $\mathbf{K}$ ASG primitives, $\mathbf{d}_i$ denotes the unit view direction pointing from the camera to each 3D Gaussian, γ is the positional encoding. This strategy enhances the ability to model complex optical phenomena compared to using SH directly for color representation. The full appearance component in Stage 2 is:

$$\mathbf{c}_{i,2} = \mathbf{c}_{i,d} + \mathbf{c}_{i,s} \quad (13)$$

The appearance loss for this stage uses the same L1 and SSIM combination as $\mathcal{L}_{s1}$ (Eq. 7), but focuses on reconstructing the appearance from the original ground truth images I_{GT} :

$$\mathcal{L}_{s2} = \mathcal{L}_{\text{RGB}}(\hat{I}, I_{GT}) \quad (14)$$

where $\hat{I}$ is the rendered image using $\mathbf{c}_{i,2}$.

Total Loss for Stage 2. The training objective for Stage 2, $\mathcal{L}_{\text{total}, s2}$, combines the appearance loss ($\mathcal{L}_{s2}$) with the same set of regularization terms used in Stage 1:

$$\mathcal{L}_{\text{total}, s2} = \mathcal{L}_{s2} + \lambda_t \mathcal{L}_{\text{trans}} + \lambda_n \mathcal{L}_n + \lambda_f \mathcal{L}_{\text{flatten}} \quad (15)$$

3.4 First-surface Depth Extraction

Gaussian Plane Depth. Inspired by PGSR [9], we utilize the concept of Gaussian plane depth which indicates the intersection depth instead of the Gaussian center depth. Following the principle of unbiased depth rendering [9] (Eq. 4), we calculate the plane depth $\hat{d}_i(\mathbf{u})$ for Gaussian $\mathcal{G}_i$ by the distance d_i , Gaussian normal $\mathbf{n}_i$ and the ray direction $\mathbf{v}_u$:

$$\hat{d}_i(\mathbf{u}) = \frac{d_i}{\mathbf{n}_i \cdot \mathbf{v}_u}. \quad (16)$$

We compute this plane depth $\hat{d}_i$ for *each individual* Gaussian. This per-Gaussian depth value then serves as the basis for our subsequent depth extraction process, rather than using a weighted average across *all* intersected Gaussians.

Limitations of Nearest Depth Approach. The naive nearest depth discussed in §1 simply takes the minimum of all the plane depths of Gaussian along the ray:

$$D_{\text{nearest}}(\mathbf{u}) = \min_i \{\hat{d}_i(\mathbf{u})\} \quad (17)$$

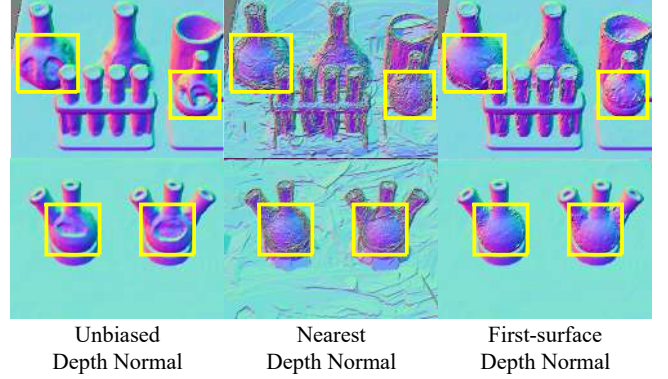


Figure 5: Comparison of normal maps derived from different depth extraction methods: unbiased depth [9], nearest depth (D_{nearest} , §3.4), our first-surface depth (D_{first} , §3.4). Our method produces significantly smoother and more accurate surface normals. In contrast, nearest depth yields noisy results due to “floaters”, while normals derived from unbiased depth can exhibit incorrect dips on transparent surfaces due to the underlying α -blending across depth.

As illustrated in Fig. 5, the nearest depth D_{nearest} exhibits a systematic underestimation bias and is heavily corrupted by “floaters” (erroneous Gaussians). This aligns with our analysis in §1 regarding the limitations of relying on individual Gaussian properties for depth extraction. To better address these issues, we need to rethink why previous works struggles to reconstruct transparent surfaces.

Challenge of Transparent Surface Reconstruction. The core challenge in transparent surface reconstruction stems from the fact that the opacity distribution required for photorealistic RGB rendering is inherently **distinct** from the one required for precise first-surface depth extraction. The former often requires lower opacities at the surface to correctly blend foreground and background light transport, while the latter ideally requires opacity concentrated sharply at the true first surface. Our approach, therefore, focuses on extracting a reliable depth-indicative distribution from the appearance-optimized opacity field learned by 3DGS, as illustrated in Fig. 3 (c).

Accumulated Transmittance Drop. For pixels predicted as transparent ($\hat{M}_T(\mathbf{u}) \geq 0.5$), our approach leverages the behavior of accumulated transmittance (AT) along the ray. As formulated in Eq. 1, T_i gradually decreases by the contribution of each Gaussian along the ray. For opaque surfaces, T_i drops sharply from nearly 1 to 0 upon hitting the surface. For transparent surfaces, this drop is typically less drastic, and T_i does not reach zero due to light penetration. Our objective is thus to robustly locate the depth corresponding to this *first significant drop* in T_i .

Maximum-Weight Window Search. We operate under the hypothesis that the region containing the true first surface will exhibit the concentration of Gaussians. To efficiently identify this region, we restrict our search along the ray using thresholds T_{start} and T_{end} . T_{start} helps to ignore potential noise or minor interactions near the camera, while T_{end} defines a point beyond which the first significant interaction is assumed to have already occurred. We employ a sliding window approach, as illustrated in Fig. 3 (c), considering

only the segment of the ray where the $T_i \in [T_{\text{end}}, T_{\text{start}}]$. This effectively targets the primary interaction zone where the first surface is most likely located.

Within this relevant ray segment $\{\mathcal{G}_i \mid T_{\text{end}} \leq T_i \leq T_{\text{start}}\}$, we define a window of a fixed size δt and slide it along the sorted candidate Gaussians. For each window position $\mathcal{W}_j$ containing a set of these candidate Gaussians, we calculate the sum of weights within it: $W_{\text{sum}}(j) = \sum_{i \in \mathcal{W}_j} T_i \alpha_i$. We then select the window $\mathcal{W}^*$ that maximizes this sum: $\mathcal{W}^* = \arg \max_j W_{\text{sum}}(j)$.

First-surface Depth Calculation. Unlike the unbiased depth approach (Eq. 4) which relies on α -blending aggregated quantities ($\mathcal{D}$ and N_w) of all Gaussians along the ray, we compute the first-surface depth $D_{\text{first}}(\mathbf{u})$ in the identified the maximum-weight window $\mathcal{W}^*$, which robustly indicates the region of the first surface interaction. We perform a weighted average of the Gaussian plane depths $\hat{d}_i(\mathbf{u})$ for Gaussians $\mathcal{G}_i$ falling within this window $\mathcal{W}^*$:

$$D_{\text{first}}(\mathbf{u}) = \frac{\sum_{i \in \mathcal{W}^*} T_i \alpha_i \hat{d}_i(\mathbf{u})}{\sum_{i \in \mathcal{W}^*} T_i \alpha_i} \quad (18)$$

Crucially, by detecting the transmittance drop to define the window $\mathcal{W}^*$ and then averaging the plane depths within it, we achieve two advantages. First, restricting the average to $\mathcal{W}^*$ effectively isolates the first surface’s contribution, preventing background details seen through transparency from corrupting the depth estimate and thus addressing the transparency-depth dilemma issue inherent in full α -blending approaches (D_{standard} , D_{unbiased}). Second, the weighted averaging within the window provides robustness against the “floaters” that plague nearest depth methods (D_{nearest}), as these noisy Gaussians typically have low $T_i \alpha_i$ or fall outside $\mathcal{W}^*$. The entire process is implemented efficiently in parallel using CUDA.

4 Experiments

4.1 Benchmark Settings

Implementation Details. Our training strategy and hyperparameters are generally consistent with 3DGS [31]. The training iterations for all scenes are set to 30,000, and 15,000 for the two stages, respectively. We adopt the densification strategy of AbsGS [78]. For mesh extraction, we begin by rendering the first-surface depth for each training view, followed by utilizing the TSDF Fusion algorithm [53] to extract the mesh. For data that does not have transparent objects, we choose not to generate a transparent mask. All our experiments are conducted on a single NVIDIA A800 GPU.

Datasets. To specifically assess performance on transparent surfaces, we introduce a new specialized dataset called **TransLab** (Transparent Laboratory). TransLab contains 8 diverse 360° scenes featuring common laboratory environments with various transparent equipment like test tubes, beakers, separatory funnels, safety goggles, flasks (including two-neck and three-neck variants), and petri dishes. This collection comprehensively represents the challenges of reconstructing transparent objects typically found in chemistry laboratories. Additionally, to evaluate general reconstruction quality, we also test our method on 15 object-centric scenes from the standard DTU dataset [27].

Evaluation Metrics. To evaluate the geometric accuracy of the reconstructed surfaces, our primary focus, we utilize the chamfer distance (CD) and F1 score. We also assess the photorealistic quality

Table 1: Quantitative comparison of geometric reconstruction quality across different methods in TransLab dataset. “Red”, “Orange”, and “Yellow” denote the best, second-best, and third-best results, respectively. Our proposed method outperforms existing mesh reconstruction techniques in terms of reconstruction quality. (§4.2)

Method	Scene								Average
	Scene 1	Scene 2	Scene 3	Scene 4	Scene 5	Scene 6	Scene 7	Scene 8	
Chamfer Distance ↓									
SuGaR [24]	37.17	10.34	8.01	6.48	8.55	46.15	14.01	7.44	17.27
2DGS [25]	2.50	8.63	2.24	2.14	4.80	2.12	2.12	3.86	3.55
GOF [79]	4.79	10.38	2.72	3.14	10.03	18.39	9.97	4.88	8.04
PGSR [9]	1.88	5.69	2.51	2.91	2.60	2.27	2.32	3.44	2.95
Ours	1.68	2.42	1.57	1.60	1.75	1.54	2.05	2.23	1.85
F1 Score ↑									
SuGaR [24]	0.25	0.49	0.59	0.56	0.57	0.26	0.34	0.46	0.44
2DGS [25]	0.91	0.55	0.90	0.89	0.69	0.98	0.97	0.84	0.84
GOF [79]	0.83	0.65	0.86	0.84	0.57	0.58	0.36	0.60	0.66
PGSR [9]	0.94	0.78	0.86	0.85	0.90	0.93	0.94	0.85	0.88
Ours	0.97	0.91	0.96	0.95	0.97	0.98	0.96	0.94	0.95

Table 2: Quantitative comparison of novel view synthesis across different methods in TransLab dataset. Our method achieves state-of-the-art results in PSNR and SSIM, and shows competitive performance in LPIPS. (§4.2)

Method	Scene								Average
	Scene 1	Scene 2	Scene 3	Scene 4	Scene 5	Scene 6	Scene 7	Scene 8	
PSNR ↑									
SuGaR [24]	23.27	23.21	22.20	21.81	22.83	23.44	24.34	23.14	23.03
2DGS [25]	38.94	30.12	38.74	37.68	34.60	36.07	41.45	37.68	36.91
GOF [79]	24.13	24.73	23.78	23.72	23.99	24.57	23.57	24.01	24.06
PGSR [9]	39.89	36.09	39.76	38.36	34.93	36.80	45.14	38.37	38.67
Ours	41.02	36.58	40.34	38.37	35.41	37.37	45.72	37.83	39.08
SSIM ↑									
SuGaR [24]	0.946	0.941	0.933	0.921	0.935	0.944	0.961	0.945	0.941
2DGS [25]	0.962	0.702	0.953	0.952	0.944	0.960	0.961	0.955	0.924
GOF [79]	0.937	0.930	0.930	0.928	0.920	0.935	0.934	0.930	0.931
PGSR [9]	0.995	0.981	0.990	0.989	0.976	0.989	0.997	0.988	0.988
Ours	0.996	0.980	0.995	0.992	0.978	0.985	0.997	0.987	0.989
LPIPS ↓									
SuGaR [24]	0.114	0.133	0.144	0.157	0.130	0.110	0.081	0.111	0.123
2DGS [25]	0.130	0.338	0.165	0.160	0.156	0.113	0.139	0.137	0.167
GOF [79]	0.123	0.137	0.132	0.133	0.144	0.118	0.130	0.131	0.131
PGSR [9]	0.010	0.030	0.020	0.019	0.027	0.018	0.007	0.015	0.018
Ours	0.010	0.038	0.013	0.015	0.032	0.025	0.006	0.019	0.020

of the rendered novel views using standard image synthesis metrics: PSNR, SSIM, and LPIPS [81].

Baseline Methods. We compared our method, TSGS, with current state-of-the-art neural surface reconstruction methods including NeuS [67], VolSDF [74], and NeuralAngelo [41]. We also compared it with recently emerged reconstruction methods based on 3DGS, such as SuGaR [24], 2DGS [25], GOF [79], and PGSR [9].

4.2 Quantitative & Qualitative Evaluation

As shown in Tab. 1, Tab. 2, and Tab. 3, our method achieves superior or highly competitive performance across different datasets.

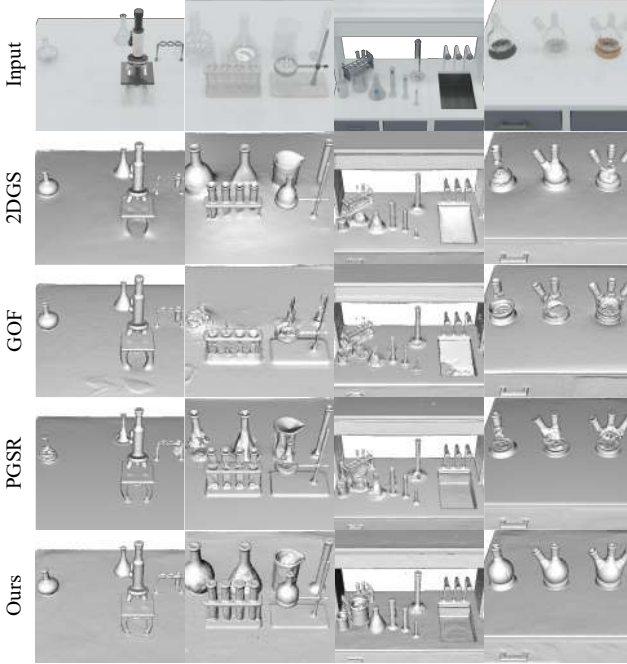
Geometric Accuracy. TSGS demonstrates superior geometric accuracy on both transparent and opaque objects. On the challenging TransLab dataset (Tab. 1), TSGS significantly improves geometric accuracy over the previous state-of-the-art (PGSR), achieving a **37.3% reduction** in average chamfer distance (1.85 vs. 2.95) and an **8.0% improvement** in average F1 score (0.95 vs. 0.88). This stems from our first-surface depth extraction and the two-stage training, which effectively resolve the transparency-depth dilemma. Furthermore, TSGS achieves state-of-the-art geometric accuracy on

Table 3: Quantitative results of chamfer distance (lower is better) on DTU dataset [27]. (§4.2)

	24	37	40	55	63	65	69	83	97	105	106	110	114	118	122	Mean
VolSDF [74]	1.14	1.26	0.81	0.49	1.25	0.70	0.72	1.29	1.18	0.70	0.66	1.08	0.42	0.61	0.55	0.86
NeuS [67]	1.00	1.37	0.93	0.43	1.10	0.65	0.57	1.48	1.09	0.83	0.52	1.20	0.35	0.49	0.54	0.84
Neuralangelo [41]	0.37	0.72	0.35	0.35	0.82	0.54	0.53	1.29	0.97	0.73	0.47	0.74	0.32	0.41	0.43	0.61
SuGaR [24]	1.47	1.33	1.13	0.61	2.25	1.71	1.15	1.63	1.62	1.07	0.79	2.45	0.98	0.88	0.79	1.33
2DGS [25]	0.48	0.91	0.39	0.39	1.01	0.83	0.81	1.36	1.27	0.76	0.70	1.40	0.40	0.76	0.52	0.80
GOF [79]	0.50	0.82	0.37	0.37	1.12	0.74	0.73	1.18	1.29	0.68	0.77	0.90	0.42	0.66	0.49	0.74
PGSR [9]	0.35	0.55	0.40	0.33	0.82	0.55	0.48	1.10	0.70	0.60	0.52	0.68	0.30	0.45	0.37	0.55
Ours	0.34	0.53	0.37	0.41	0.79	0.55	0.48	1.05	0.62	0.59	0.42	0.49	0.31	0.37	0.35	0.51

Table 4: Ablation study. We evaluate the effectiveness of components in our method on both geometry and appearance metrics. (§4.3)

Setting	Geometry		Appearance		
	CD↓	F1↑	PSNR↑	SSIM↑	LPIPS↓
Full Model	1.85	0.95	39.08	0.989	0.020
w/o ASG	1.86	0.96	37.54	0.987	0.023
w/o Normal Prior	1.96	0.94	38.84	0.989	0.019
w/o De-light Prior	1.87	0.95	38.95	0.989	0.020
w/o First-surface Depth	1.89	0.95	39.08	0.989	0.020
w/o Fixing Opacity Params	1.87	0.95	38.60	0.988	0.021

**Figure 6: Comparison with other methods on the TransLab dataset. Our method accurately reconstructs transparent surfaces, while other methods fail to reconstruct the correct geometry, often resulting in holes or dips on transparent regions. (§4.2)**

the standard opaque DTU dataset [27] (Tab. 3), demonstrating the robustness and generalizability of our geometry learning approach. **Novel View Synthesis.** Our method also yields high visual quality on TransLab (Tab. 2). Compared to PGSR, TSGS improves average PSNR by 0.41dB (39.08 vs. 38.67) and slightly improves average SSIM (0.989 vs. 0.988). Our LPIPS (0.020) is comparable to the baseline (0.018). This suggests our staged approach achieves a good balance between geometric accuracy and visual quality.

Qualitative Results. Fig. 6 demonstrates TSGS’s capability in transparent object reconstruction. Compared to baselines, our method

better preserves fine geometric details and achieves more realistic renderings. Due to the limited space, more qualitative results are shown in Appendix.

4.3 Ablation Study

We ablate key components of TSGS, and results are in Tab. 4.

Normal Prior. Excluding the Normal prior [75] from Stage 1 (§3.2) most significantly impacts geometry (CD: 1.85 vs. 1.96). This highlights the critical need for strong geometric priors to guide reconstruction for ill-posed transparent surfaces.

De-light Prior. Removing the De-light prior [75] from Stage 1 (§3.2) slightly worsens geometry (CD: 1.85 vs. 1.87) and PSNR (39.08 vs. 38.95dB). The De-light prior helps handle strong highlights during initial geometry learning.

First-surface Depth Extraction. Replacing our first-surface depth extraction (§3.4) with unbiased depth extraction formulated in Eq. 4 slightly increases CD (1.85 vs. 1.89). While rendered appearance metrics are unaffected, this step is vital for accurate first-surface geometry.

Anisotropic Spherical Gaussians. Replacing ASG (§3.3) with standard Gaussians significantly degrades appearance (PSNR: 39.08 vs. 37.54dB), demonstrating ASG’s importance for modeling view-dependent specularities. Geometry is minimally affected as ASG primarily refines appearance in Stage 2.

Fixing Opacity Parameters. Optimizing all parameters jointly in Stage 2, instead of freezing opacity (§3.3), degrades both appearance (PSNR: 39.08 vs. 38.60dB) and geometry. This validates our two-stage strategy, where fixing opacity allows focused appearance refinement without compromising geometric accuracy.

5 Conclusion

Reconstructing transparent surfaces using 3D Gaussian Splatting is hampered by the transparency-depth dilemma, where appearance optimization conflicts with accurate geometry extraction. We introduced TSGS, a framework designed to resolve this dilemma. TSGS utilizes a two-stage training strategy to decouple geometry learning from appearance refinement. During inference, a novel first-surface depth extraction method robustly identifies the surface using a sliding window over rendering weights. Evaluated on our challenging new TransLab dataset, TSGS demonstrates state-of-the-art performance, significantly improving geometric accuracy (37.3% chamfer distance reduction, 8.0% F1 score improvement) over strong baselines while improving visual quality (+0.41dB PSNR). By effectively decoupling geometry and appearance and employing robust depth extraction, TSGS enables simultaneous high-fidelity reconstruction and rendering of transparent objects within the 3DGS framework.

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A Appendix

A.1 TransLab Dataset

The rise of embodied AI in automated laboratories necessitates accurate 3D reconstruction and manipulation of transparent glassware. To address the lack of specialized benchmarks for this challenging task, the TransLab dataset provides a comprehensive synthetic benchmark for evaluating reconstruction algorithms on transparent objects. It features a diverse collection of 8 scenes in total, containing typical laboratory glassware, including test tubes, beakers, safety goggles, flasks (two-neck and three-neck variants), petri dishes, graduated cylinders, round-bottom flasks, Erlenmeyer flasks, and condensers.

For each scene, we provide extensive ground truth data, generated using Blender’s physically-based rendering (PBR) engine, to facilitate thorough evaluation. This includes high-resolution RGB imagery (I_{GT}) rendered at an original resolution of 1600x1600 pixels (downscaled to 800x800 for training), reference 3D ground truth meshes (M_{GT}), background segmentation masks (M_{BG}), masks identifying transparent objects (M_T), dense depth maps (D), surface normal maps (N), and environmental illumination maps (E). A summary of these data types is also provided in Table 5. This rich set of annotations enables detailed analysis of reconstruction quality from multiple perspectives.

Table 5: Ground Truth Data Provided in the TransLab Dataset

Symbol	Description
I_{GT}	High-resolution RGB imagery
M_{GT}	Reference 3D ground truth meshes
M_{BG}	Background segmentation masks
M_T	Transparent object segmentation masks
D	Dense depth maps
N	Surface normal maps
E	Environmental illumination maps

A.2 Quantitative Evaluation Details

To quantitatively assess reconstruction performance, we employ established metrics for both image quality and geometric fidelity. Image quality is measured using PSNR, SSIM, and LPIPS. Geometric accuracy is evaluated using two primary metrics:

Chamfer Distance (CD). This metric quantifies the average distance between the predicted mesh surface M_{pred} and the ground truth mesh surface M_{GT} . To compute the CD, we first uniformly sample a dense set of points P_{pred} from the surface of the predicted mesh M_{pred} and similarly sample a point set P_{GT} from the ground truth mesh M_{GT} . The chamfer distance is then calculated as the symmetric average of the mean shortest distances between the points in these two point clouds.

Geometry F1 Score. This metric assesses the balance between the precision and recall of the reconstructed mesh vertices. Let V_{pred} be the set of vertices in the predicted mesh M_{pred} and V_{GT} be the set of vertices in the ground truth mesh M_{GT} . A predicted vertex $v \in V_{pred}$ is considered a true positive (TP) if its Euclidean distance

to the nearest ground truth vertex $v' \in V_{GT}$ is less than a predefined threshold $\tau = 0.005$:

$$TP = \{v \in V_{pred} \mid \min_{v' \in V_{GT}} \|v - v'\|_2 < \tau\} \quad (19)$$

Precision (P) is defined as the ratio of true positives to the total number of predicted vertices, $P = |TP|/|V_{pred}|$. Recall (R) is defined as the ratio of true positives to the total number of ground truth vertices, $R = |TP|/|V_{GT}|$. The F1 score is the harmonic mean of precision and recall:

$$F1 = 2 \cdot \frac{P \cdot R}{P + R} \quad (20)$$

A.3 Error Analysis of Depth Extraction

We set the depth window size δt to 3 mm in our maximum-weight window search. Since the final estimated depth d_{first} is computed as a weighted average within the selected window $\mathcal{W}^*$, the theoretical depth error is naturally bounded by the window size, i.e., at most $\delta t = 3\text{mm}$.

A.4 Limitations and Future Work

Limitations. Our framework currently assumes single-layer transparency; the first-surface depth extraction may be less accurate for complex multi-layer refractive phenomena (e.g., liquids in containers) leading to caustics. Additionally, modeling highly complex or spatially varying anisotropic material appearances remains an avenue for improvement.

Future work. Future work can directly address these limitations. Key directions include: (1) Extending our first-surface extraction method to handle multi-layer transparencies and complex refractive effects more accurately, potentially by integrating principles from physically-based ray tracing. (2) Improving the modeling of complex, spatially varying material appearances by leveraging advanced neural BRDF representations or integrating material spectroscopy priors. These advancements promise to further enhance the accuracy and realism of transparent object reconstruction.

A.5 Additional Qualitative Results

In this section, we present additional qualitative results to further demonstrate the performance of our method, TSGS, on both the challenging TransLab dataset and the general DTU benchmark.

A.5.1 TransLab Dataset Results. Fig. 7 and Fig. 8 showcases more reconstruction examples from the TransLab dataset. These results highlight TSGS’s ability to handle diverse transparent laboratory equipment with varying geometric complexity and material properties. Our method consistently recovers accurate first-surface geometry while rendering photorealistic appearances, effectively addressing the challenges posed by transparency.

A.5.2 DTU Dataset Results. To illustrate the generalizability of our approach, Fig. 9, Fig. 10, Fig. 11, Fig. 12, Fig. 13 present additional qualitative results on selected scenes from the DTU dataset. Although optimized for transparent surfaces, TSGS demonstrates strong performance on these standard opaque objects, achieving high-fidelity surface reconstruction comparable to state-of-the-art methods designed for general scenes. This underscores the robustness of our geometry learning stage. For the results of PGSR, 2DGS,

and GOF, we directly adopt the visualizations from the original PGSR paper, please note that minor viewpoint discrepancies may exist as we did not have access to their original camera and projection parameters.

A.6 Additional Quantitative Results

A.6.1 Results on Real-world Dataset. To validate our method’s performance in real-world scenarios, we conduct a quantitative evaluation on the highly challenging ClearPose [12] dataset. This benchmark features 63 scenes with highly transparent objects, which often exhibit complex overlaps and out-of-focus blur. For this evaluation, we selected “scene1” from each of the four sets. Given the high density of views in the original dataset, we subsampled the training data by selecting one image every 100 frames. The quantitative results are presented in Table 6. Note that the ClearPose dataset provides ground truth (GT) only for the target objects by aligning a mesh with the object in the captured images. Consequently, GT is not available for other parts of the scene, such as the table or background. Thus, for geometric evaluation, we compute a unidirectional Chamfer Distance from the predicted mesh to the GT object mesh.

A.6.2 Computational Efficiency. As shown in Table 7, our 3DGS-based method is significantly faster than NeRF-based approaches and exhibits performance comparable to other 3DGS-based methods. Specifically, our method’s training time of 0.8 hours is substantially lower than the 13 hours required by NU-NeRF. It is worth

noting that due to its requirement for NVIDIA OptiX, the NU-NeRF results were obtained on a machine equipped with an NVIDIA RTX 4090 GPU. In terms of rendering speed, our approach achieves 105 FPS, which is on par with 2DGS and faster than PGSR. This efficient performance, combined with moderate VRAM usage, underscores our method’s suitability for applications requiring rapid 3D reconstruction.

A.6.3 Comparison with Baselines Using Normal Priors. To further evaluate our method, we provide a comparison against baseline methods enhanced with the same normal priors on the TransLab dataset. As shown in Table 8, our approach consistently outperforms the enhanced baselines across key metrics. This demonstrates that the superior performance of our method is attributed to its fundamental design, rather than being solely dependent on the auxiliary normal information.

A.6.4 Comparison with NU-NeRF on the TransLab Dataset. We also conducted a direct comparison against NU-NeRF on the TransLab dataset. A significant challenge observed with NU-NeRF is its tendency to suffer from convergence issues on this dataset, often resulting in degenerated reconstructions that collapse into a simple spherical shape. In contrast, our method demonstrates robust convergence and superior reconstruction quality. The quantitative comparison of the average metrics is presented in Table 9, which clearly highlights the performance gap between the two methods.

Table 6: Quantitative evaluation on the ClearPose [12] dataset. For each scene, we compare our method against 2DGS [25], PGSR [9], and NU-NeRF [62]. Best results are in bold.

Scene	Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	CD $\downarrow$	F1 $\uparrow$
<i>Chemical</i>						
Set 1	2DGS [25]	21.20	70.6	0.447	30.03	5.35
	PGSR [9]	21.18	78.0	0.384	15.28	22.37
	NU-NeRF [62]	20.53	73.3	0.440	27.88	7.45
	Ours	22.20	77.3	0.348	7.73	27.56
<i>Household</i>						
Set 2	2DGS [25]	16.85	40.8	0.499	36.15	8.22
	PGSR [9]	13.39	27.0	0.574	20.92	7.00
	NU-NeRF [62]	16.99	49.7	0.503	12.02	36.44
	Ours	17.90	54.0	0.410	7.48	36.53
Set 3	2DGS [25]	15.30	40.7	0.524	45.77	17.10
	PGSR [9]	13.65	25.7	0.561	13.88	4.06
	NU-NeRF [62]	16.23	53.6	0.501	10.55	37.47
	Ours	16.89	54.4	0.412	6.54	42.09
Set 4	2DGS [25]	14.50	37.8	0.543	54.48	12.67
	PGSR [9]	14.52	29.3	0.580	11.36	24.06
	NU-NeRF [62]	17.37	62.8	0.386	17.35	35.13
	Ours	15.90	57.4	0.393	7.89	36.86

Table 7: Computational efficiency comparison. Our method is significantly faster than NeRF-based approaches and comparable to other 3DGS-based methods.

Method	Training Time (hour)	VRAM (GB)	Render FPS
2DGS [25]	0.2	2	108
PGSR [9]	0.5	6	87
NU-NeRF [62]	13	20	0.03
Ours	0.8	7.7	105

Table 8: Comparison on the TransLab dataset with baseline methods enhanced with normal priors. Best results are in bold.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	CD $\downarrow$	F1 $\uparrow$
2DGS+Normal	29.68	97.7	0.040	2.09	92.6
PGSR+Normal	38.22	98.6	0.020	2.20	91.2
Ours	39.08	98.9	0.020	1.85	95.0

Table 9: Quantitative comparison with NU-NeRF on the TransLab dataset. We report the average metrics across all scenes. Best results are in bold.

Method	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	CD $\downarrow$	F1 $\uparrow$
NU-NeRF	28.20	92.9	0.144	34.68	36.0
Ours	39.08	98.9	0.020	1.85	95.0

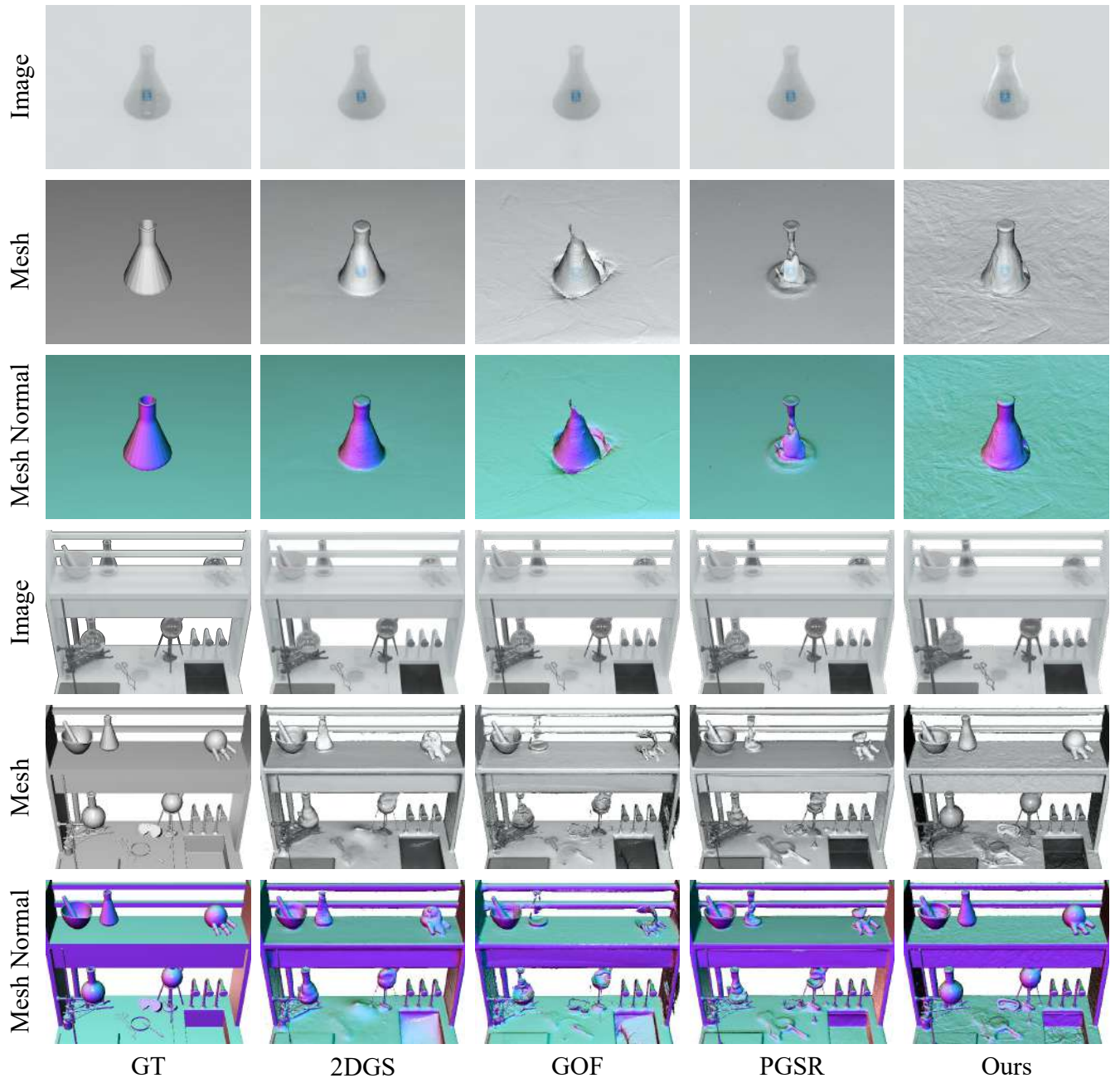


Figure 7: Additional qualitative reconstruction results on various scenes from the TransLab dataset. Our method (TSGS) successfully captures the geometry and appearance of complex transparent objects like test tubes, beakers, and flasks.

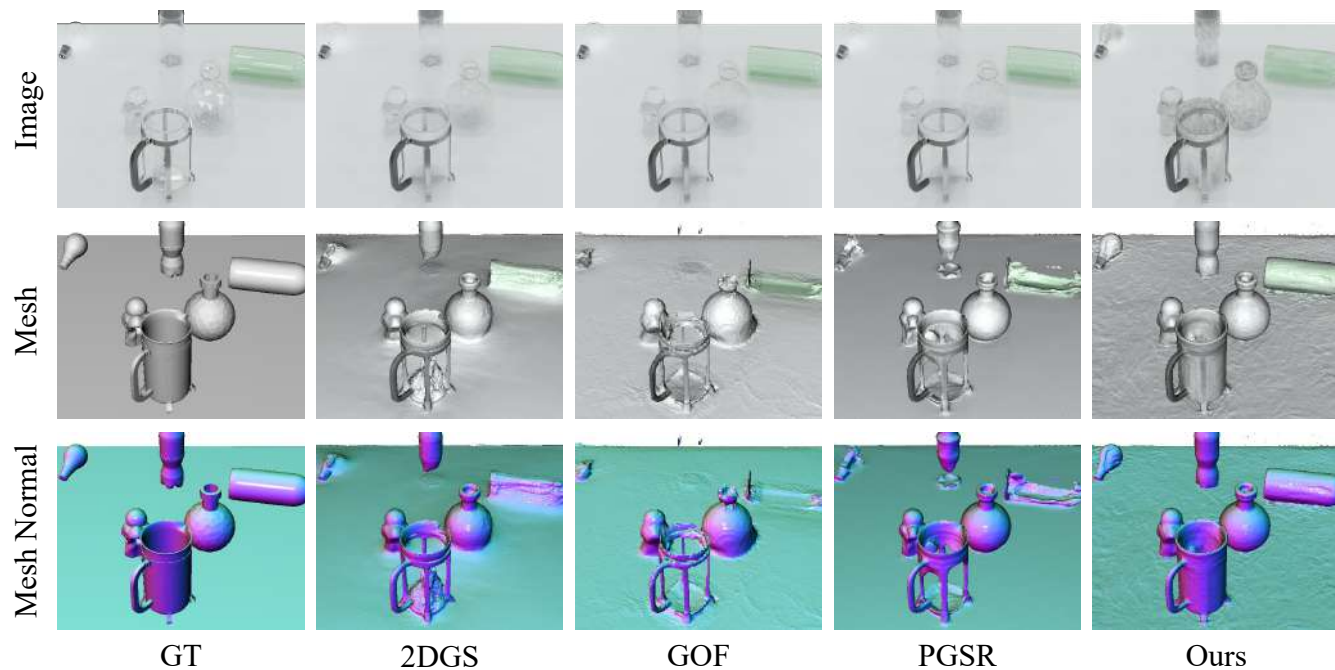


Figure 8: Additional qualitative reconstruction results on various scenes from the TransLab dataset. Our method (TSGS) successfully captures the geometry and appearance of complex transparent objects like test tubes, beakers, and flasks.

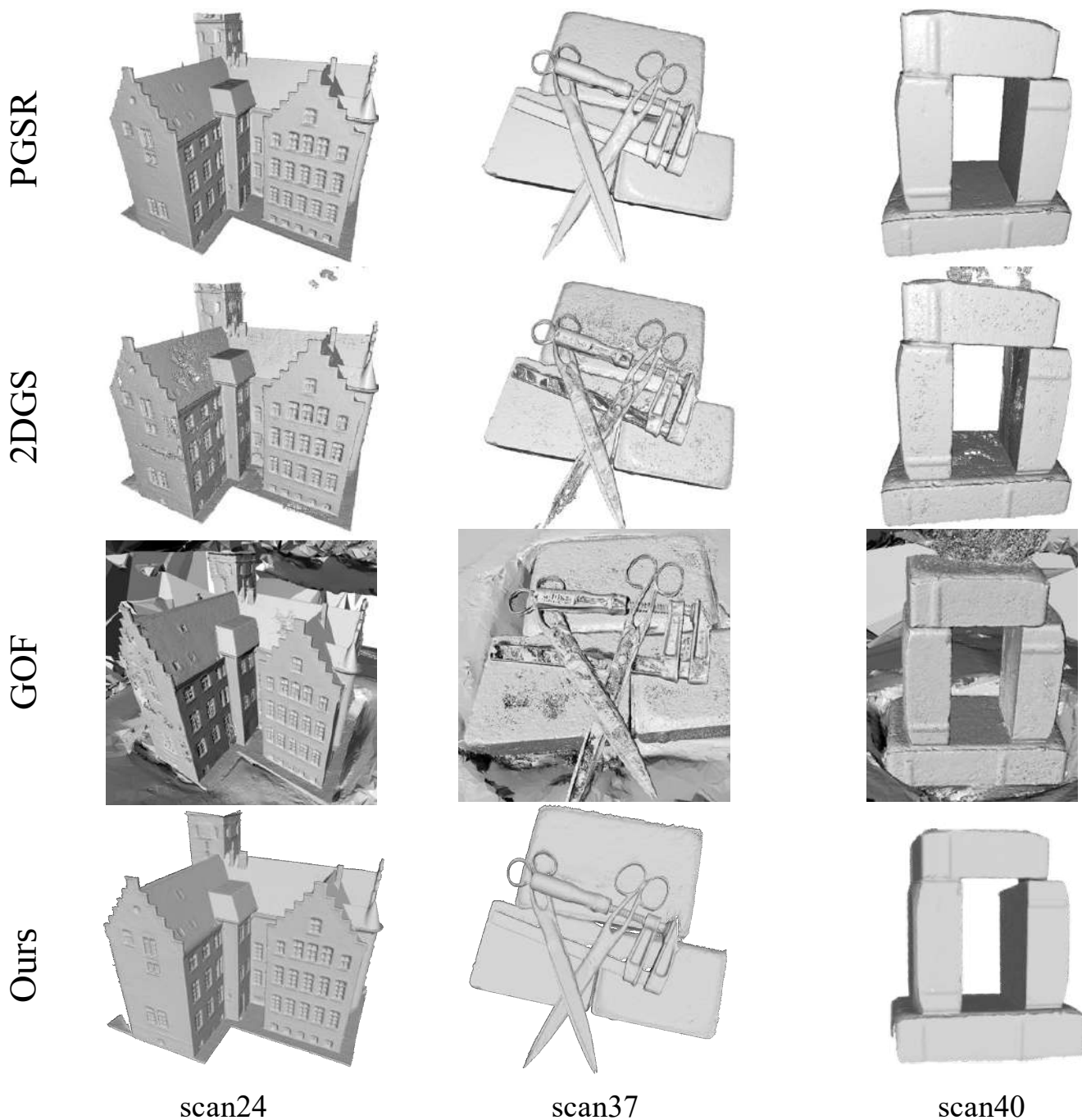


Figure 9: Additional qualitative reconstruction results on scenes from the DTU dataset. This demonstrates the effectiveness of our method (TSGS) on general opaque objects, achieving competitive reconstruction quality.

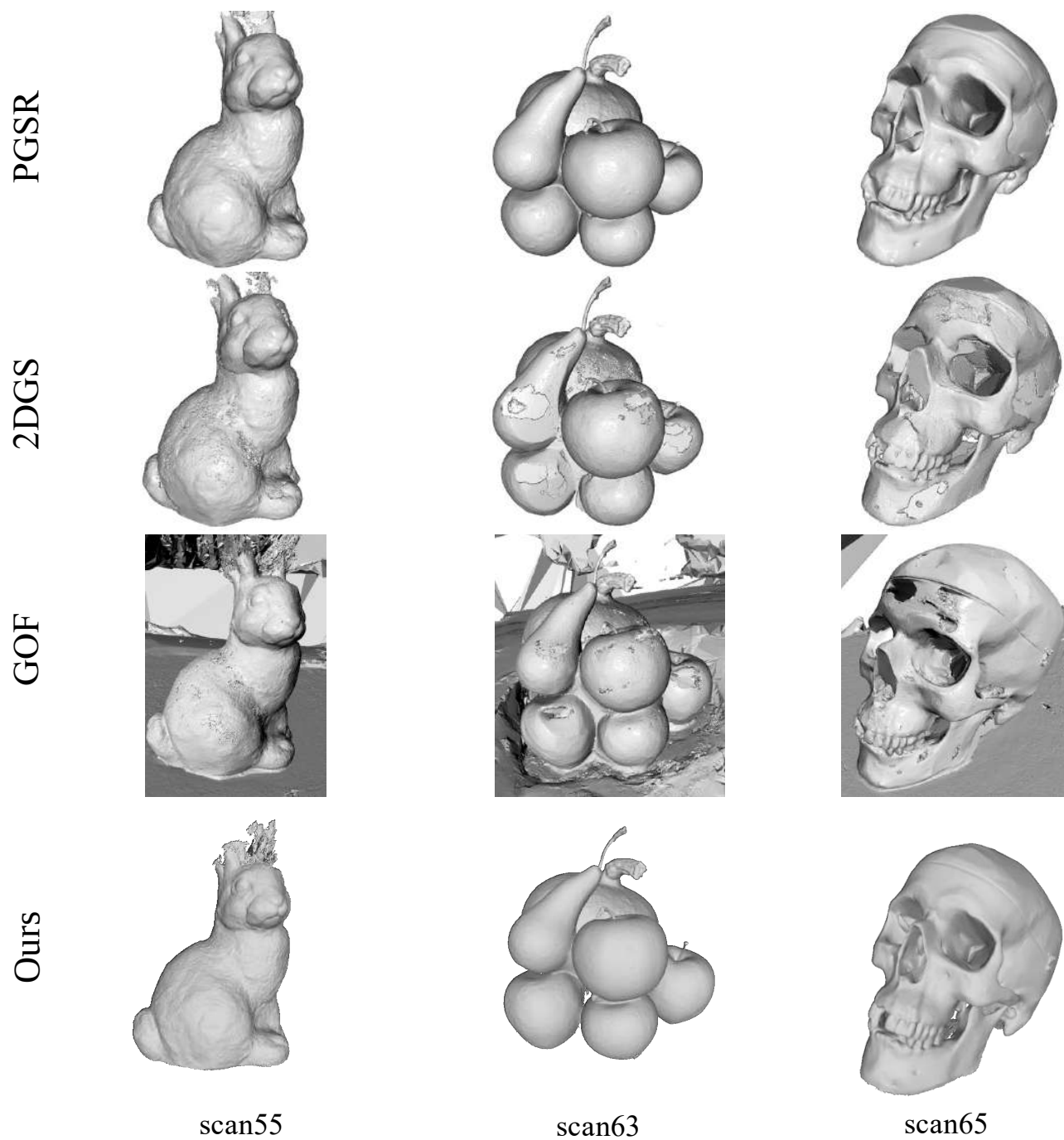


Figure 10: Additional qualitative reconstruction results on scenes from the DTU dataset (Continued). This demonstrates the effectiveness of our method (TSGS) on general opaque objects, achieving competitive reconstruction quality.

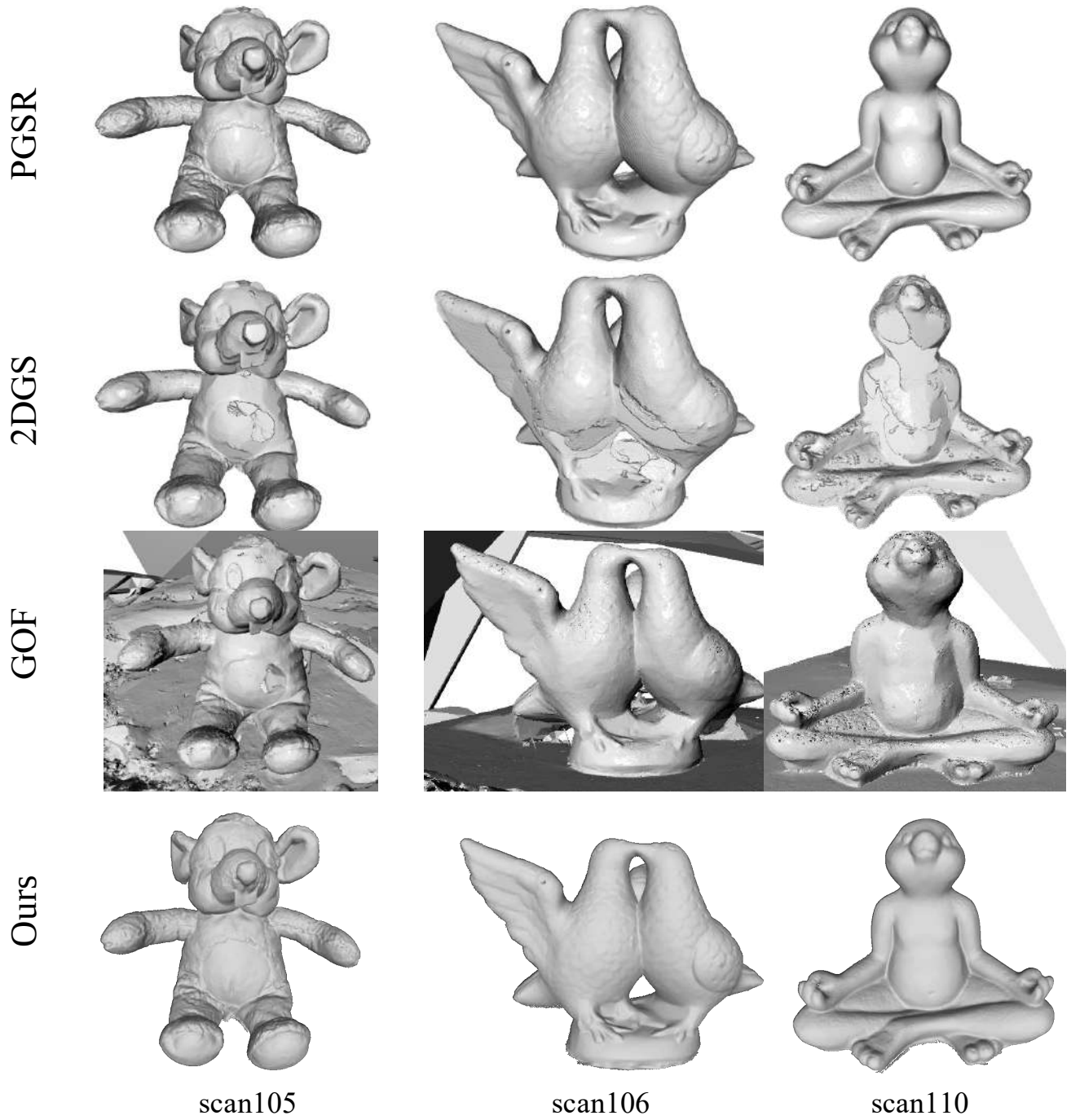


Figure 11: Additional qualitative reconstruction results on scenes from the DTU dataset (Continued). This demonstrates the effectiveness of our method (TSGS) on general opaque objects, achieving competitive reconstruction quality.

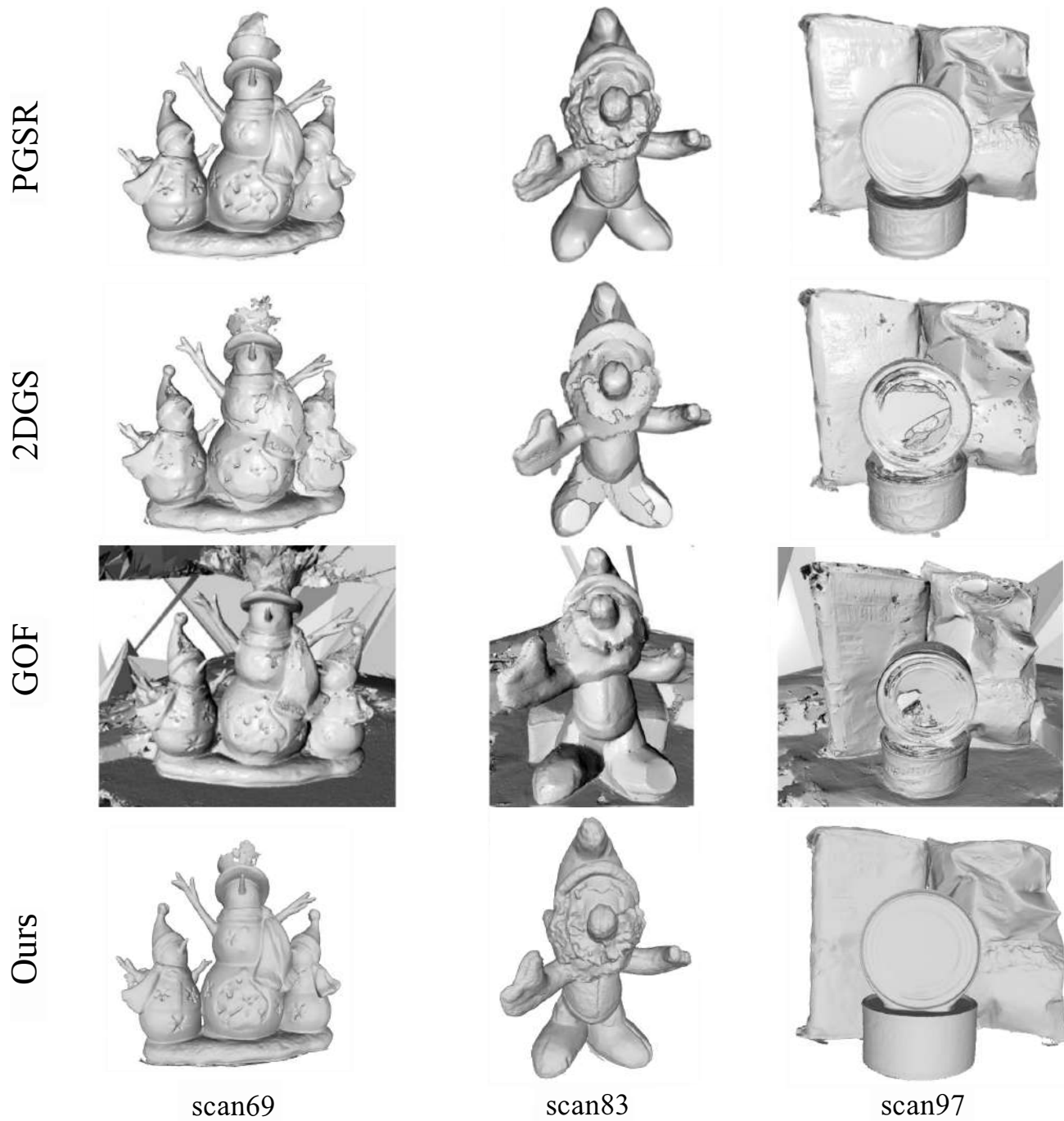


Figure 12: Additional qualitative reconstruction results on scenes from the DTU dataset (Continued). This demonstrates the effectiveness of our method (TSGS) on general opaque objects, achieving competitive reconstruction quality.

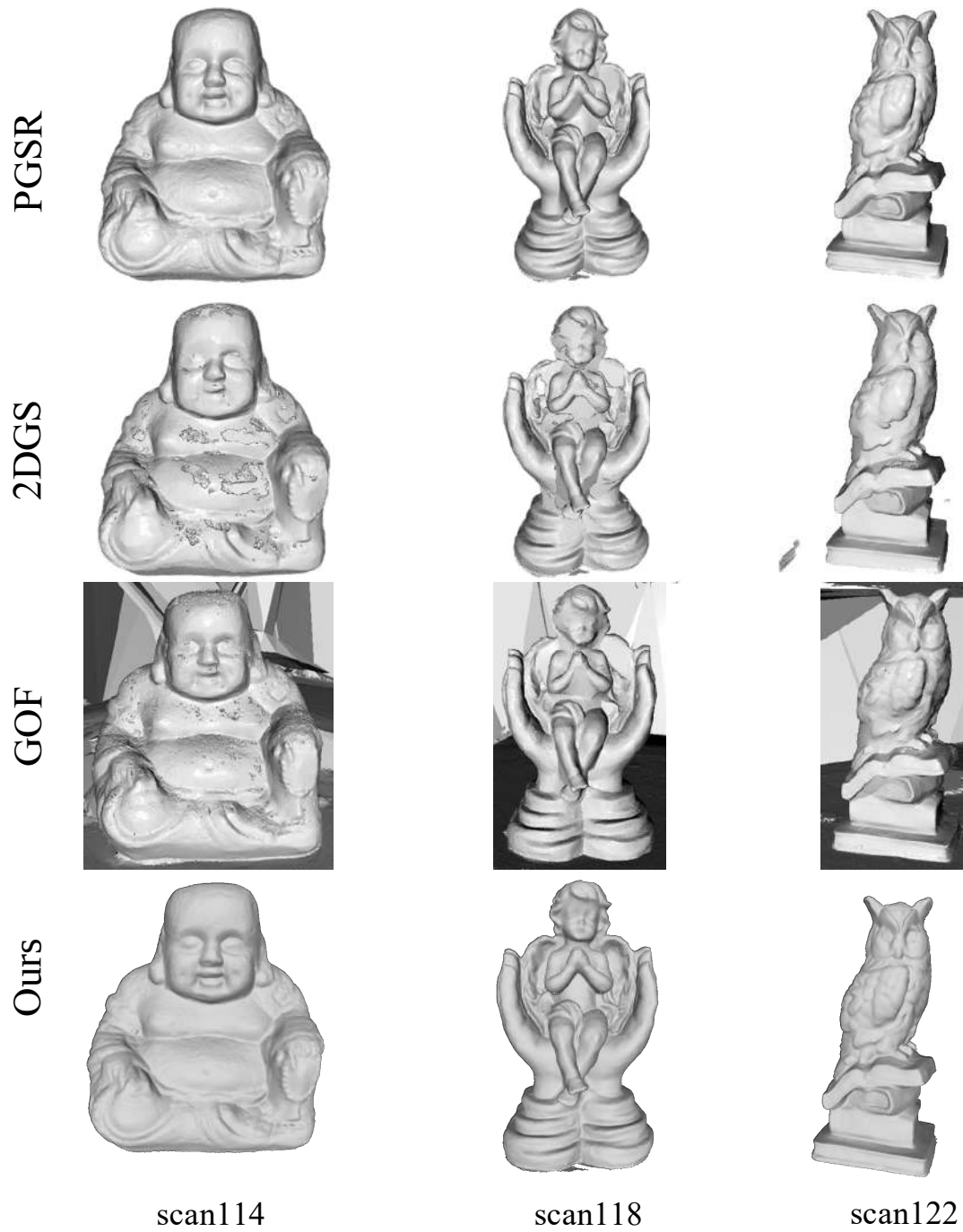


Figure 13: Additional qualitative reconstruction results on scenes from the DTU dataset (Continued). This demonstrates the effectiveness of our method (TSGS) on general opaque objects, achieving competitive reconstruction quality.